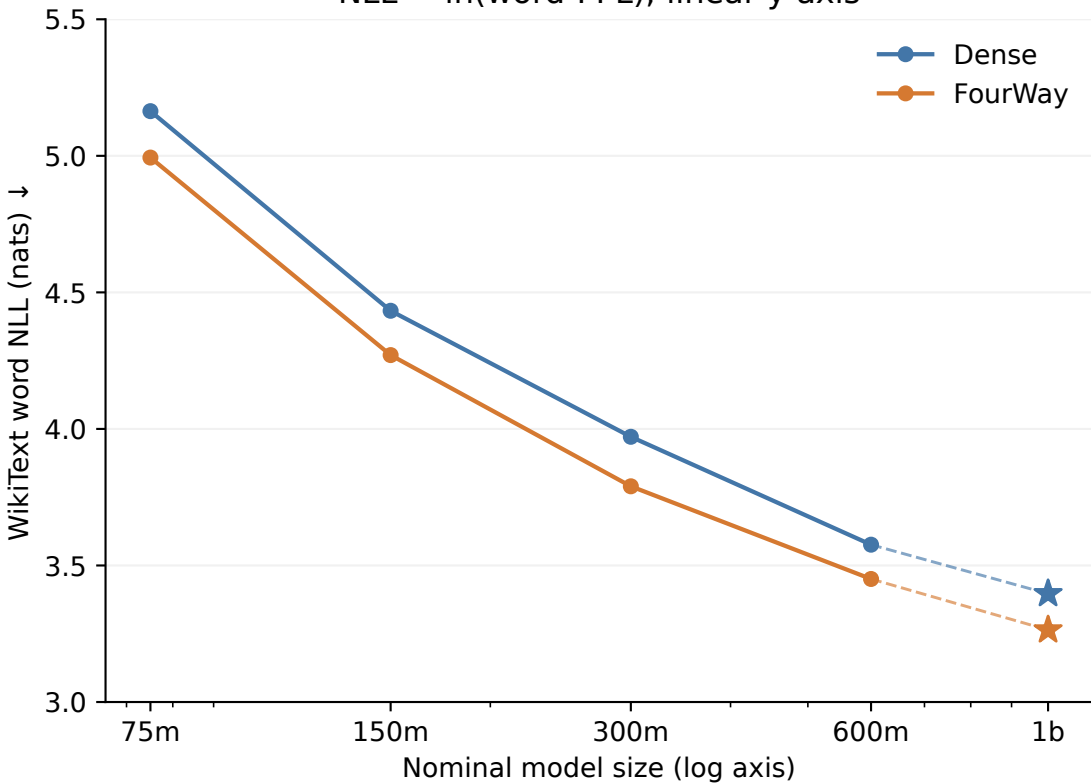


Descriptive NLL trends; entropy floor is not estimated
300M: 4.72B training tokens; ★ 1B: OLMo-mix, other sizes: Dolma

NLL = $\ln(\text{word-PPL})$; linear y axis



Log-log fit: $\text{NLL} = A \cdot N^{-\alpha}$, with $E = 0$

